

VECTO 0.x



VECTO v0.11.4 Development Release (02-04-2025)

Features

- CodeEU #855: IEPC with multiple load curves (!321)
- H2 ICE vehicles in declaration mode (!325)

Fixes

- Add max-windows-size to fuel cell (!312)
- Electric system power demand compensation for FCS (!313)
- Add error message for unknown completed vehicle missions (!330)
- Segment lookup method (!331)
- FullLoadCurves proper initialization (!332)
- Manage 'GetTruckSegment' exception behavior (!333)

VECTO 0.11.2.3456 Development Version 18.06.2024

Release Notes

- Features
 - CodeEU-196: Implement wheel bearings
 - Input for Engineering mode is available via the GUI & JSON file. In the vehicle file (.vveh) in an entry of the "Axles" array, you can add the field "WheelEndFriction" and assign it a numerical value. Input for Declaration mode is available via a temporary XSD (VectoDeclarationDefinitions.DEV.2.6.xsd). Sample file in folder: Generic Vehicles\Declaration Mode\Wheel_Bearings
 - CodeEU-643: added support for fuel cell IEPC2
 - Added new simulation job type for IEPC vehicles with fuel cells. This feature is only supported in engineering mode. The GUI has been amended to allow the creation/modification of this new job type. A sample job has been added to the folder 'Generic Vehicles\Engineering Mode\GenericIEPC - FCHV'.

- Bugfixes
 - CodeEU-236: Fuel Cell Simulation Aborts when Charging Power of Battery is too low
 - CodeEU-209: VECTO-0.11.1.3228-DEV crash in declaration mode
 - CodeEU-207: Writing XML Reports fails for vehicles without battery

VECTO 0.11.1.3382 Development Version 03.11.2023

Release Notes

- Changes
 - Enabled the usage of multiple Fuel Cell Systems (FCS)
 - Up to two FCS Strings
 - Up to three FCS in each FCS String
 - New naming convention:
 - FCS: Single Fuel Cell System, described with MassFlowMap (.vfcm) and power limits container in the Fuel Cell System File (.vfcc).
 - FCS String: Defined in the .vveh file, can contain up to three identical FCS
 - CFCS: Composite Fuel Cell System, a virtual fuel cell system containing up to two FCS String.
 - Changes in input files due to new naming convention
 - Optimized power distribution among FCS Strings and between individual FCS inside a string. (Explained in more detail in the user manual)
 - Allow operating a FCS below its minimum power via time splitting
 - Removed On/Off - Hysteresis
 - Removed Gradient Power Change

First VECTO beta release for the 3rd Amendment H2 Internal Combustion Engines

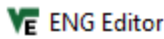
VECTO 0.11.0.3193 Development Version 29.09.2023

Release Notes

H2 ICE

New fuel types implemented "H2 PI" and "H2 CI" (i.e. H2-Diesel dual-fuel).

Well-known ICE input data structure unchanged, no new elements





Make and Model

Idling Engine Speed [rpm]
 Rated Speed [rpm]

Displacement [ccm]
 Rated Power [kW]

Dual Fuel Engine ☒
 Max Torque [Nm]

Inertia incl. Flywheel [kgm²]

Full Load and Drag Curve



Primary Fuel
 Secondary Fuel

Fuel Type

Diesel CI
 Ethanol CI
 Petrol PI
 Ethanol PI
 LPG PI
 NG PI
 NG CI
 H2 PI
 H2 CI

Fuel Consum



Fuel Consum

Correction Fa

WHTC Urban
 WHTC Motorway

Cold/Hot Emission Balancing Factor
 Reg. Per. Corr. F.

Engineering

WHR Type

☐ Mechanical WHR System included
 ☐ Mechanical WHR System not con
 ☐ Electrical WHR System

Fuel characteristics used

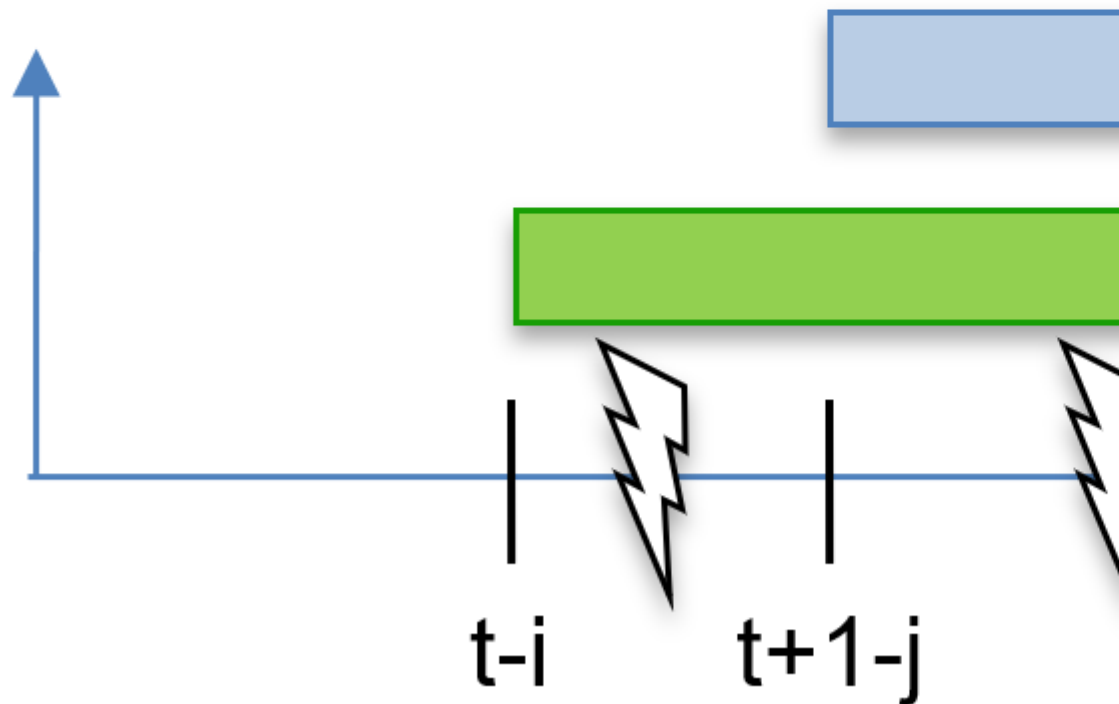
FuelType	FuelDensity [kg/m3]	CO2 per FuelWeight [kgCo2/kgFuel]	CO2 per FuelWeight VTP [kgCO2/kgFuel]	NCV_stdEngine [kJ/kg]	NCV_stdVecto [kJ/kg]
H2 PI	<empty>	0.00	0.00	120000	120000
H2 CI	<empty>	0.00	0.00	120000	120000

First VECTO beta release for the 3rd Amendment Fuel Cell Hybrid Vehicles (FCHV)

Fuel cell hybrid vehicles (FCHV) - Op. strategy (1/5)

- FCHV generic operation strategy (Charge Sustaining Mode) implemented as presented in TF FCS #15 with following basic principle:
 - Maximum phlegmatisation of FCS by operation strategy
 - FCS needs to provide average electric power demand over longer horizon
 - Longest possible averaging horizon depends on battery size and specific vehicle-cycle combination ☑ _ The smaller the battery or the greater the electric energy demand of the vehicle, _ _ the smaller the averaging horizon will get leading to a more dynamic operation of the FCS.
- Basic approach for implementation of FCS operation in “CS” Mode in 4 steps
 - Pre-simulation run as PEV to determine electric power demand for propulsion and auxiliaries
 - Power limit of “special” battery defined as sum of respective maximum powers (BAT + FCS)
 - Resulting total electric power from step 1 will be averaged over a certain *fixed* window size which is distance based (details next slide)
 - Buffering of energy is limited by battery size, SOC limits min/max are not allowed to be exceeded
 - Using largest possible window size without violating SOC limits in any window
 - Adaption of start SOC to specific mission in order to maximize usage of battery as buffer
 - Actual simulation run as FCHV in-the-loop where FCS is operated following a fixed power trace to cover average electric power demand determined from pre-processing in step 2 (battery covers instantaneous deviations from average electric power demand)
 - Δ SOC correction in post-processing to account for small deviations from neutral SOC behaviour over cycle (e.g. variations in auxiliary power, small deviations in battery losses etc.)
- Averaging and window size explained (1/2)
 - The averaging window is placed distance-symmetrically over each actual time step from the pre-simulation

- Window size is determined by total distance within window



- Averaging and window size explained (2/2)
 - Maximum window size is given by complete cycle, thus cycle is concatenated 3 times in a row
 - For time steps near the beginning or end of the cycle, the windows expand over the actual cycle and continue at the adjacent parts at the end or beginning.



- Actual simulation run as FCHV and applicable boundary conditions
 - In case average power demand over total cycle is higher than FCS maximum power: defined error message and simulation is aborted (vehicle configuration not reasonable for this mission, should not occur in practice)
 - In case average power in a certain timestep is lower than FCS minimum power: FCS is switched off missing energy from FCS for those timesteps will be accounted for in post-processing via ΔSOC correction

Fuel cell hybrid vehicles (FCHV) - VECTO features (1/5)

New powertrain type of FCHV added to Job Editor

Tools ▾ Help ▾

Job Files (

- Job Editor - Conventional Vehicle
- Job Editor - Serial Hybrid Vehicle
- Job Editor - Parallel Hybrid Vehicle
- Job Editor - Battery Electric Vehicle
- Job Editor - IEPC-E Vehicle
- Job Editor - IEPC-S Vehicle
- Job Editor - IHPC Vehicle
- Job Editor - Fuel Cell Vehicle
- Job Editor - Engine Only Mode
- VTP Job Editor
- Vehicle Editor
- Engine Editor
- Gearbox Editor
- Graph

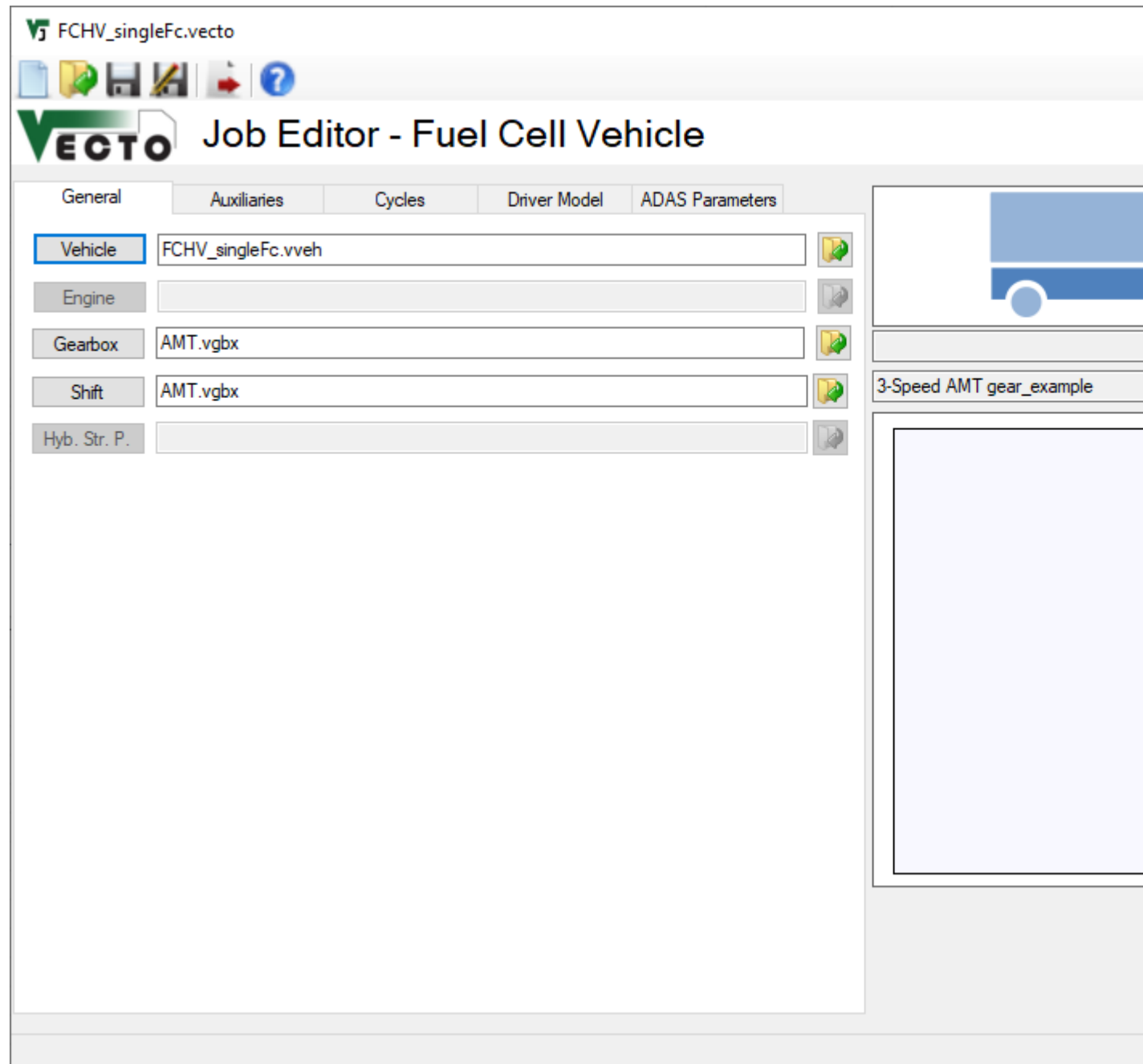
Open Log

Settings

+ - ☐ Select All ▲ ▼

Message

FCS to be parameterized via specific “Fuel Cell System” tab in “Vehicle” editor



VEH Editor

VECTO Fuel Cell Vehicle

Rigid Truck 4x2

Technically Permissible Maximum Laden Mass 39.99 [t]

HDV Group - Unkn

General Powertrain Electric Machine REESS Fuel Cell System ADAS PTO

Fuel Cell System

Gradient Power Change 30 [kW/s]

On/Off-Hysteresis 10 [s]

Fuel Cell Component	Count
GenericFuelCellComponent_100kW.vfcc	1

+ - (Double-Click to Edit)

Save Cancel

Unsaved changes in current file

Power gradient limits change in FCS power over time

On/off-Hysteresis is minimum time that FCS stays off or on after state change occurs ☒ _ not required anymore after insights gained during work on implementation of multiple FCS_

FCS to be parameterized via specific "Fuel Cell System" tab in "Vehicle" editor

VEH Editor



 **Fuel Cell Vehicle**

Rigid Truck

4x2

Technically Permissible
Maximum Laden Mass

39.99 [t]

HDV Group

-

Unkn

General

Powertrain

Electric Machine

REESS

Fuel Cell System

ADAS

PTO

Fuel Cell System

Gradient Power Change

30 [kW/s]

On/Off-Hysteresis

10 [s]

Fuel Cell Component	Count	
GenericFuelCellComponent_100kW.vfcc	1	

+

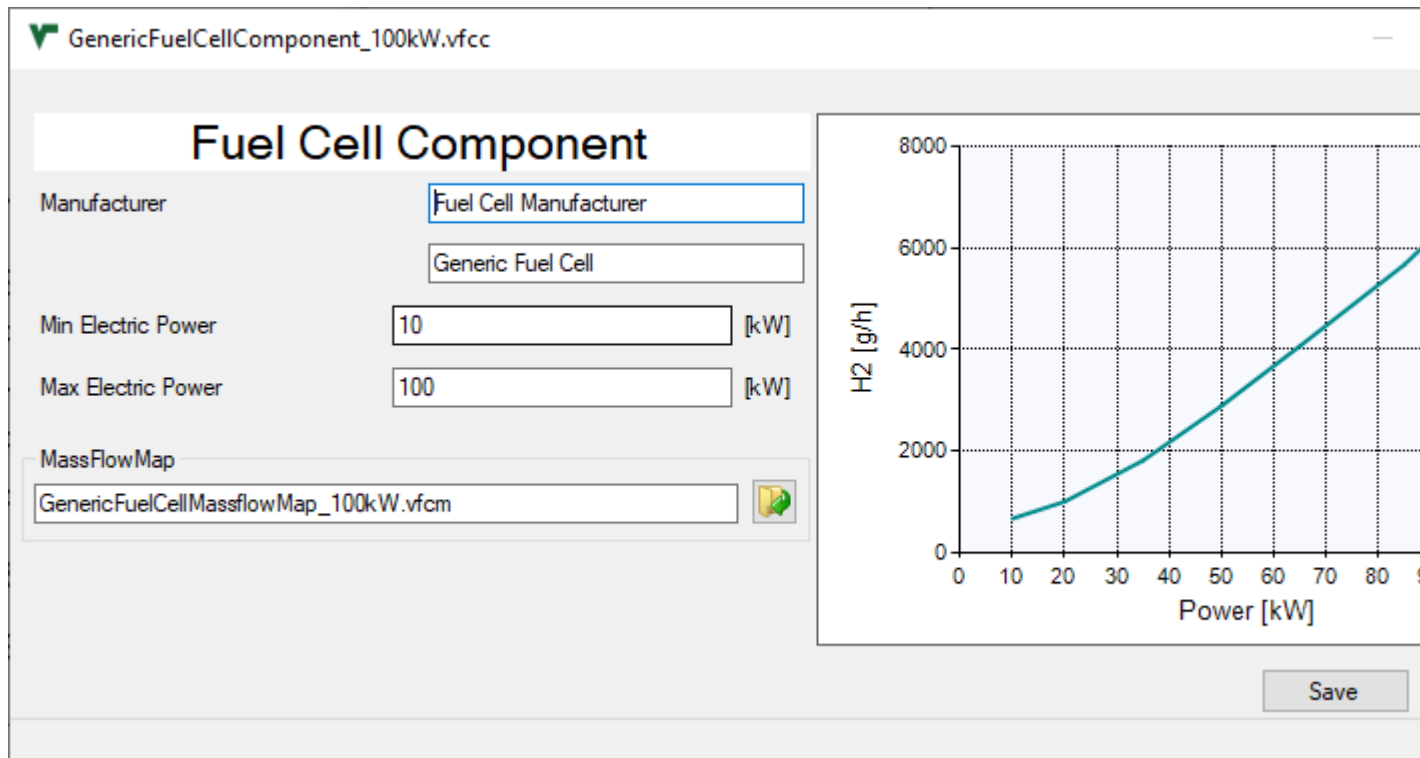
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(Double-Click to Edit)

Save

Cancel

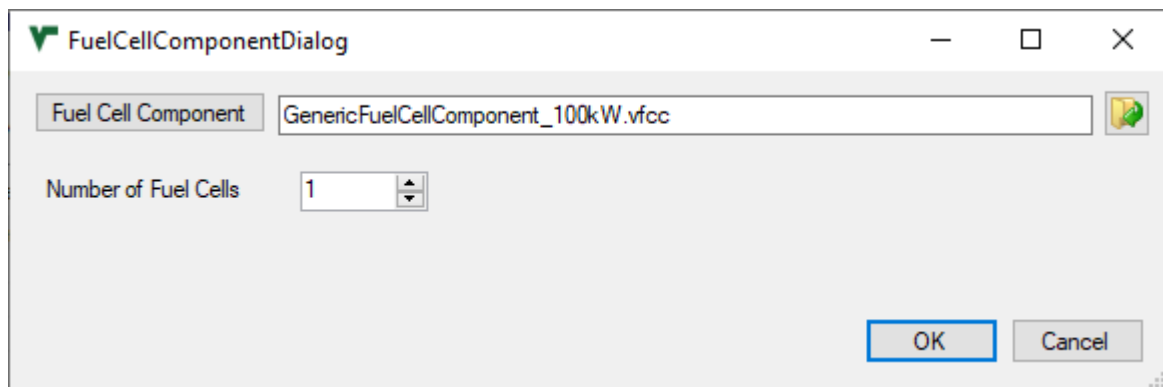
Unsaved changes in current file



Max / min power limits operation range of FCS component

FCS mass flow map needs to cover the indicated range

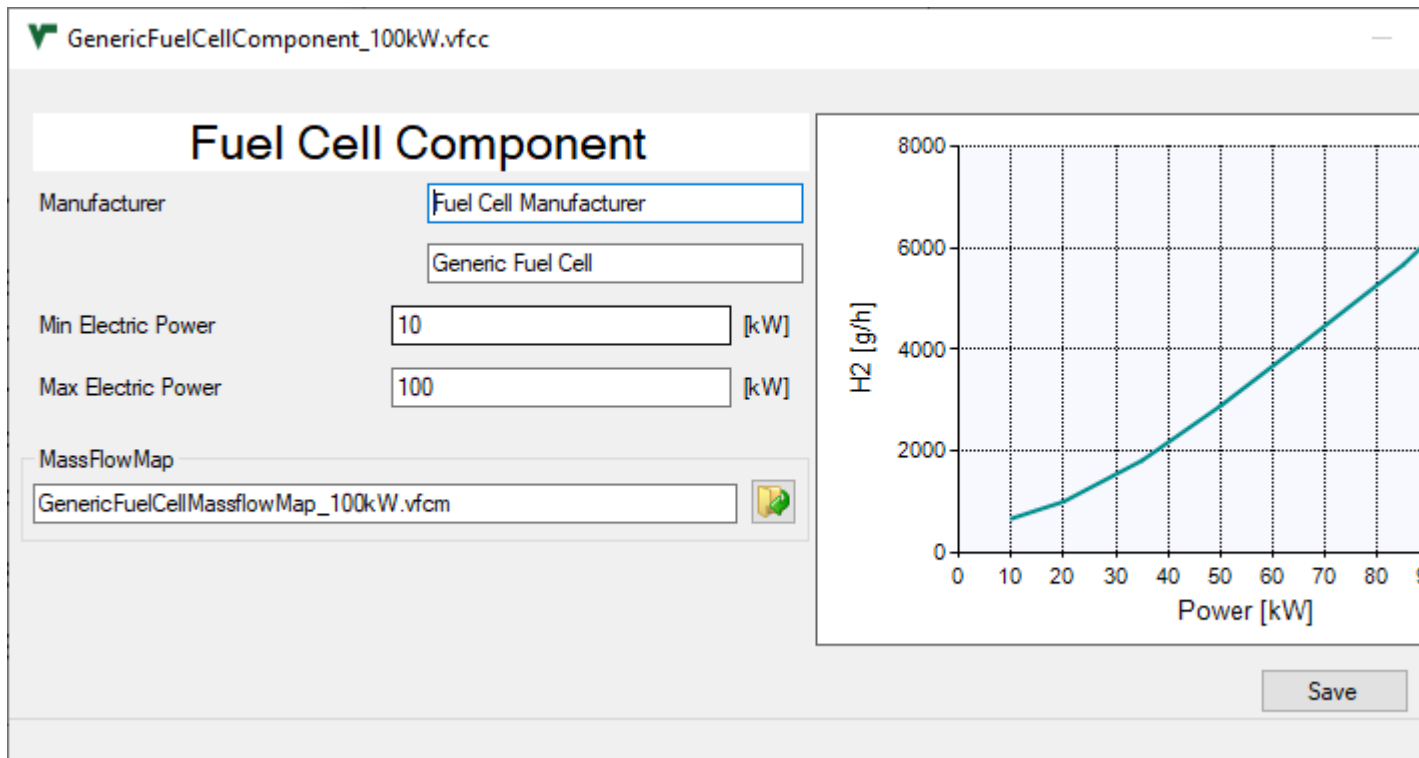
Double-click or “+”



Limited to 1

for this release

FCS to be parameterized via specific “Fuel Cell System” tab in “Vehicle” editor



- Input data structure for FCS H2 mass flow map
- CSV file format
 - 1 st column: Electric power output of FCS in [kW]
 - 2 nd column: H2 mass flow in [g/h]
- *FCS component efficiency should also include DC/DC converter (as this is part of component test procedure)!*

P_el_out [kW],	m_H2 [g/h]
10.0,	1200.0
30.0,	2002.0
60.0,	3003.0
105.0,	5436.5
150.0,	8662.5
195.0,	12199.7
255.0,	17017.0
300.0,	21450.0

- Parameterization of auxiliaries for FCHV
 - Electric power demand of auxiliaries for both, lorries and buses, need to be parameterized via the "Bus Auxiliaries" GUI window

INFO: Electric aux power demand is calculated by multiplying the specified current values by a voltage of 28.3V



Bus Auxiliaries Engineering

Electric System

Current Demand Engine On		[A]
Current Demand Engine Off Driving		[A]
Current Demand Engine Off Standstill		[A]
Alternator Efficiency		[-]
Alternator Technology	Conventional	
Max Recuperation Power		[W]
Useable Electric Storage Capacity		[Wh]
Electric Storage Efficiency		[-]
☒ ES supply from HEV REESS		
DC/DC Converter Efficiency		[-]

Pneumatic System

Compressor Map		
Average Air Demand		[NI/s]
Compressor Ratio		[-]
☐ Smart Air Compressor		

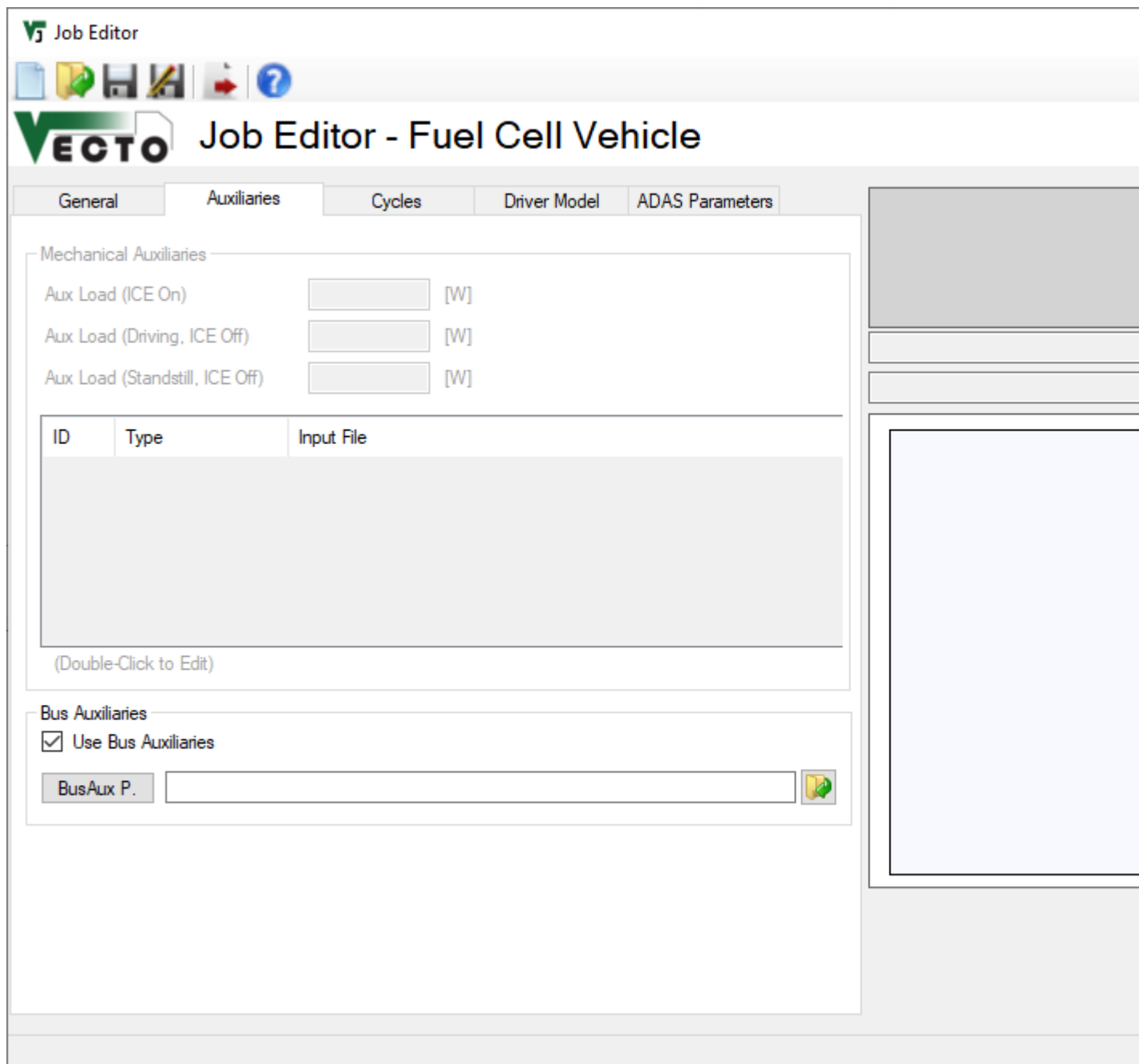
HVAC S

Mecha

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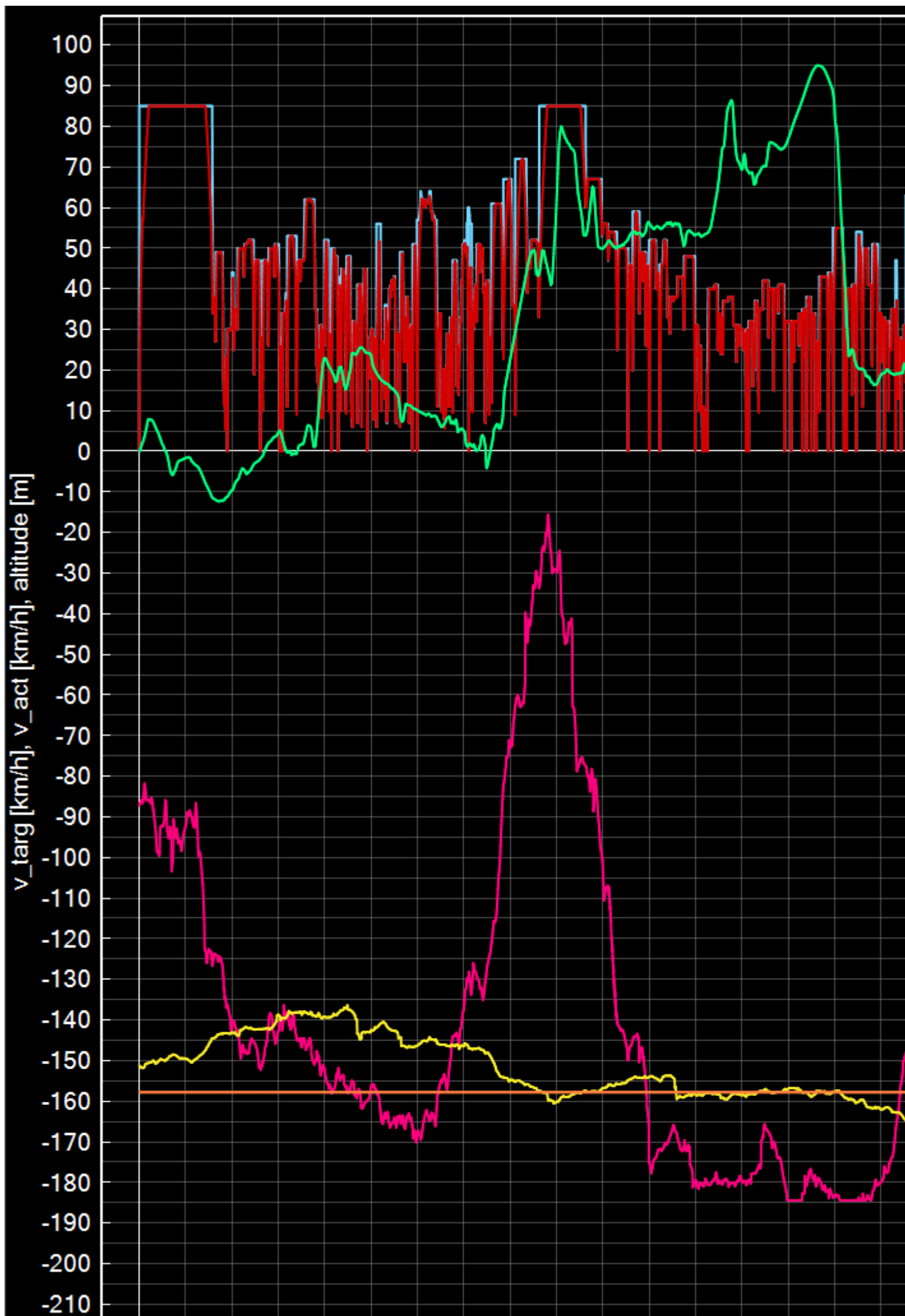
Demand during vehicle driving

Demand during vehicle standstill

FCHV - Examples for generic operation strategy (1/5)

- Exemplary results for VECTO Group 2 12-ton rigid lorry (9.5 ton actual mass) on following slides
- Simulation in VECTO Urban Delivery Cycle
 - Average electric power demand around 22 kW
- FCHV powertrain specifications:
 - 100 kW FCS max power
 - following variants of battery:
 - 5 kWh
 - 10 kWh

- 20 kWh
- Resulting speed profile was exactly the same for all different configurations



- The lower the buffer size, the more dynamic the FCS operation gets
- *Exactly same speed profile for all variants _ _*
- Averaging window sizes determined:
 - 5 kWh: 6 km
 - 10 kWh: 37.5 km
 - 20 kWh: 100 km

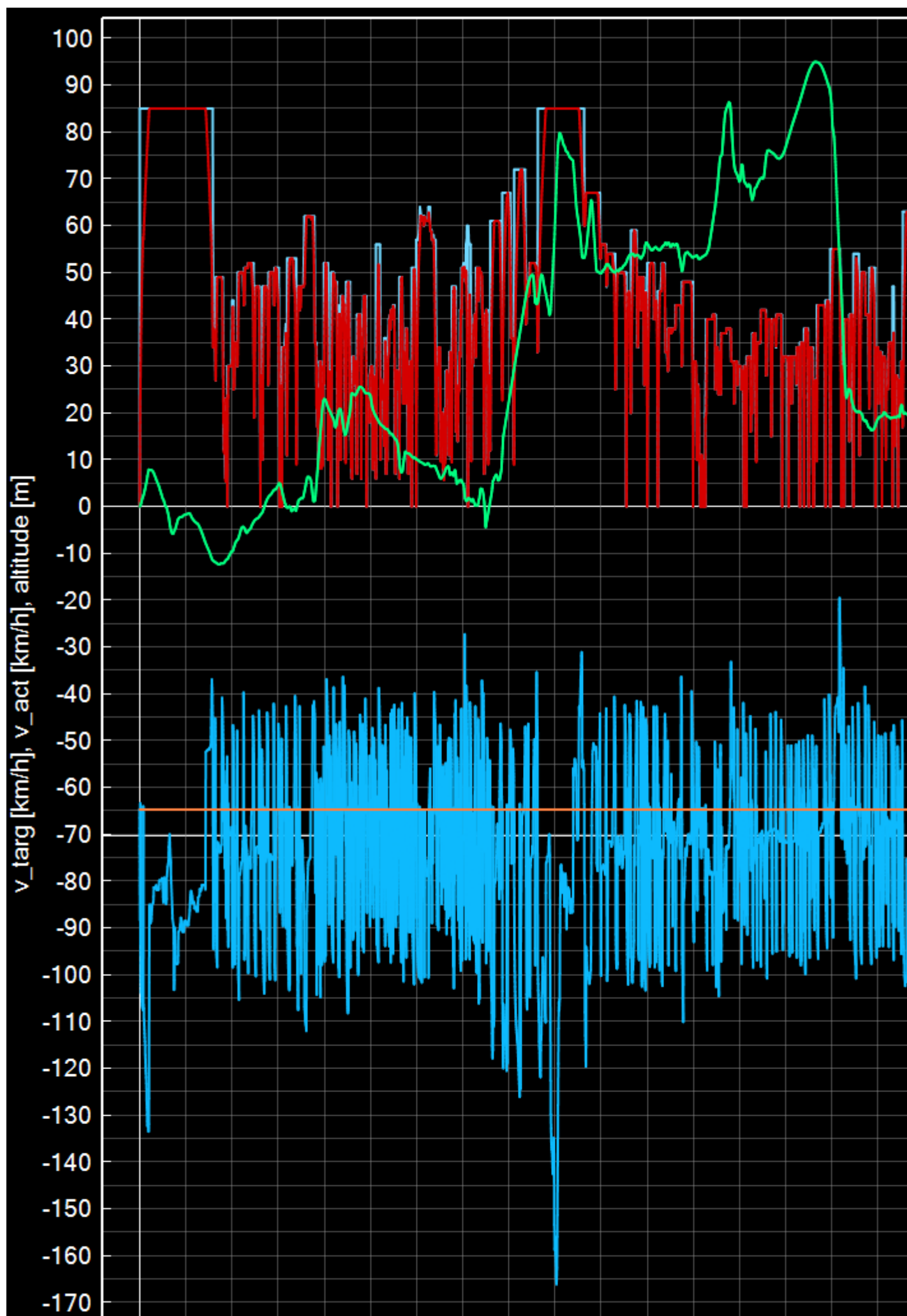
[A] v_targ [km/h]	L
[A] v_act [km/h]	L
[A] altitude [m]	L
[C] P_FuelCellSystem_actual [kW]	R
[A] P_FuelCellSystem_actual [kW]	R
[B] P_FuelCellSystem_actual [kW]	R

5 kWh

10 kWh

20 kWh

Fuel cell power for different battery sizes



Battery covers instantaneous deviations from average electric power demand

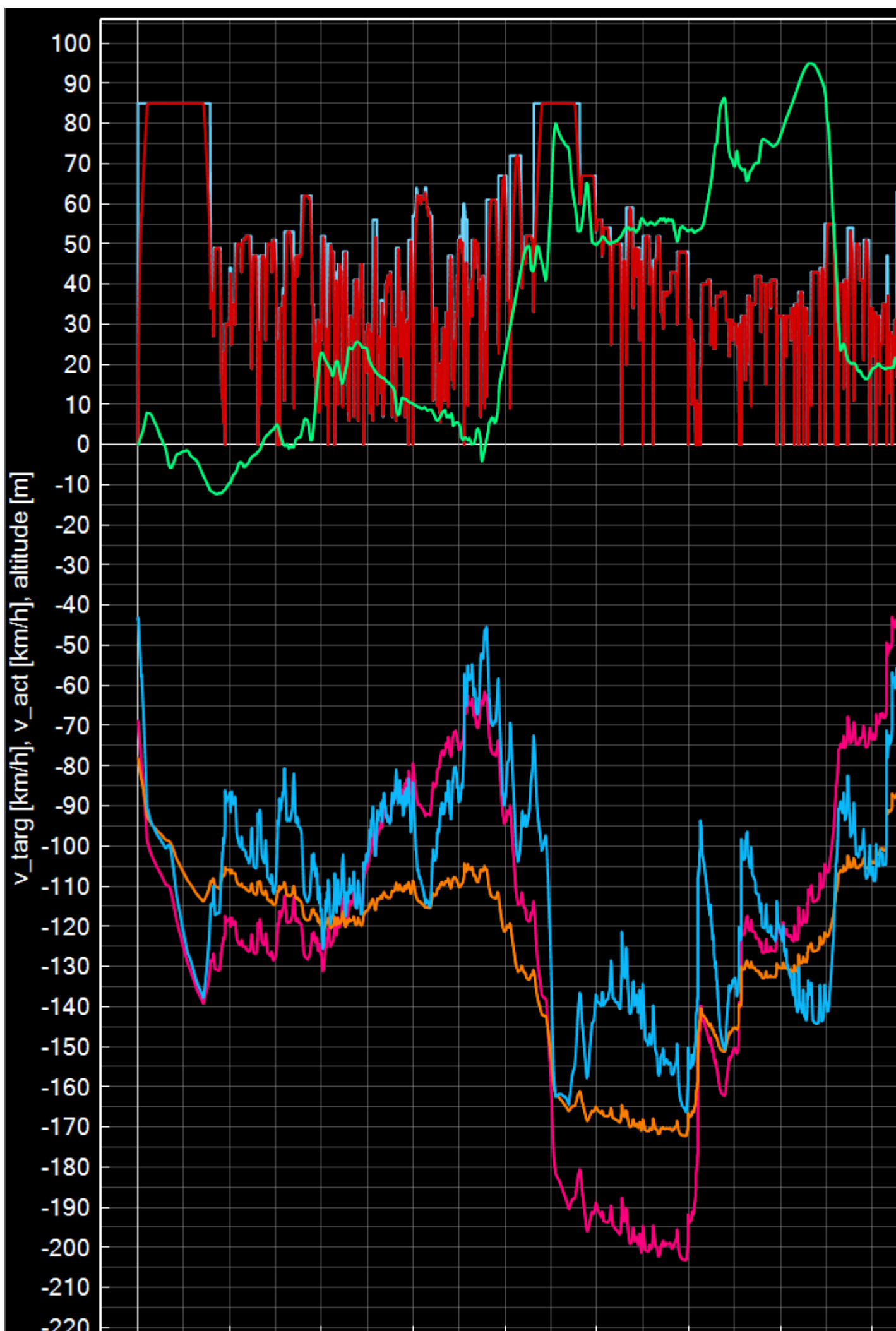
Average electric power demand (i.e. FCS power) constant for the setting shown with 20 kWh REESS size _

—

[A] v_targ [km/h]	L
[A] v_act [km/h]	L
[A] altitude [m]	L
[B] P_REESS_ES_T [kW] / dist	R
[B] P_FuelCellSystem_actual [kW]	R

Battery power

Fuel cell power



Using largest possible window size without violating SOC limits in any window ☒ maximizing of battery buffer usage

Exactly same speed profile for all variants _ _

SOC limits are Engineering settings, no connex to generic limits used in Declaration Mode

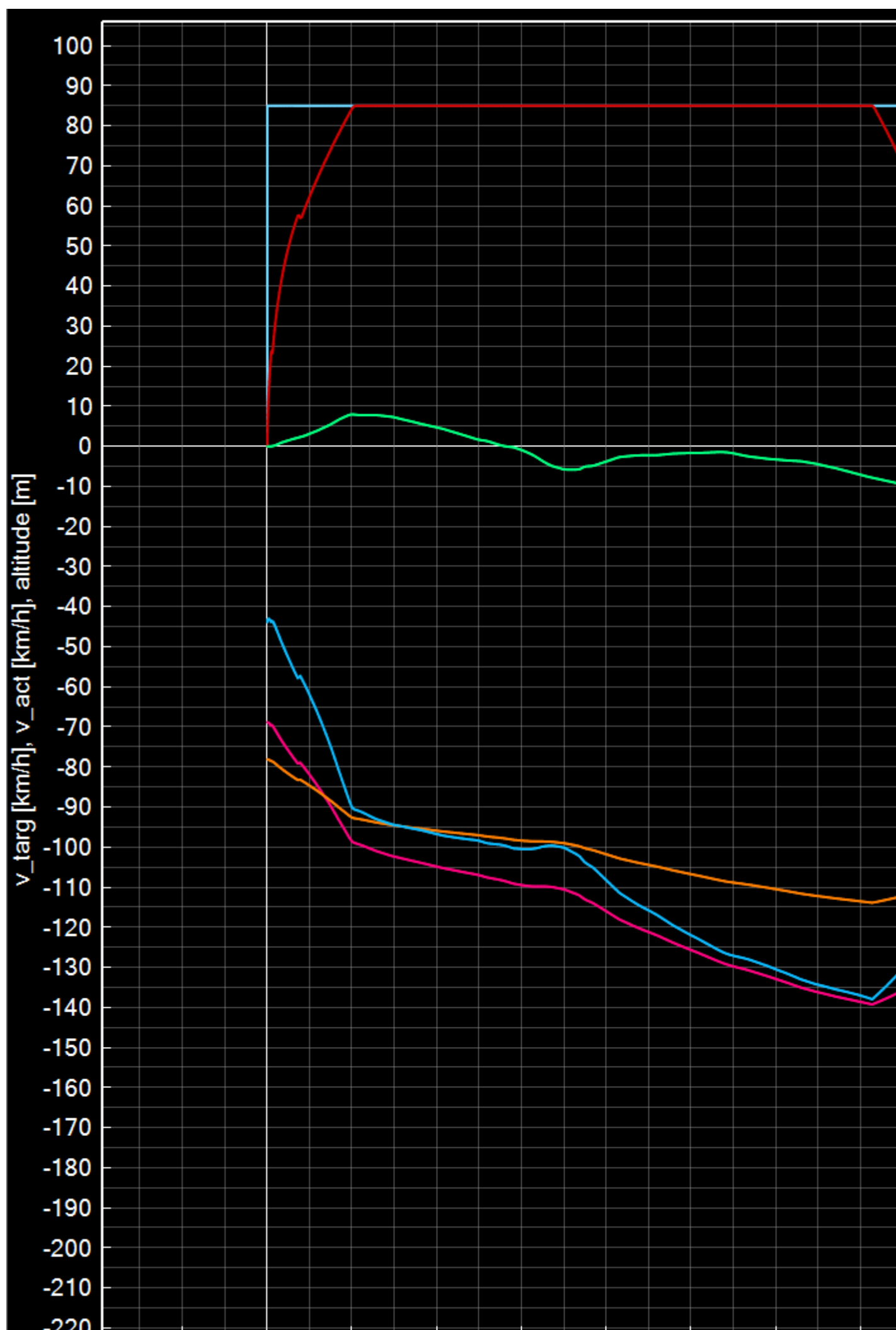
[A] v_targ [km/h]	L
[A] v_act [km/h]	L
[A] altitude [m]	L
[A] REESS SOC [%]	R
[B] REESS SOC [%] / dist [m]	R
[C] REESS SOC [%]	R

SOC for different battery sizes

5 kWh

10 kWh

20 kWh



Adaption of start SOC to specific mission in order to maximize usage of battery as buffer (detail from previous graph)

Exactly same speed profile for all variants

Adapted start SOC

[A] v_targ [km/h]	L
[A] v_act [km/h]	L
[A] altitude [m]	L
[A] REESS SOC [%]	R
[B] REESS SOC [%] / dist [m]	R
[C] REESS SOC [%]	R

SOC for different battery sizes: * 5 kWh * 10 kWh * 20 kWh

Fuel cell hybrid vehicles (FCHV) - Comments

- Release to be distributed via VECTO-DEV mailing list end of September
 - Basic description of FCHV features will also be added to VECTO release notes
 - Some example vehicles will be provided
- Current release restricted to one single FCS component
- **Updated release end of October will be able to handle:**
 - multiple FCS configurations as presented in TF FCS #17
 - Maximum 2x3 configuration i.e. 2 different FCS with a maximum number of 3 identical systems for each of the 2 different ones
 - Special provisions for handling power demands below minimum FCS power
 - IMC feature combined with FCHV

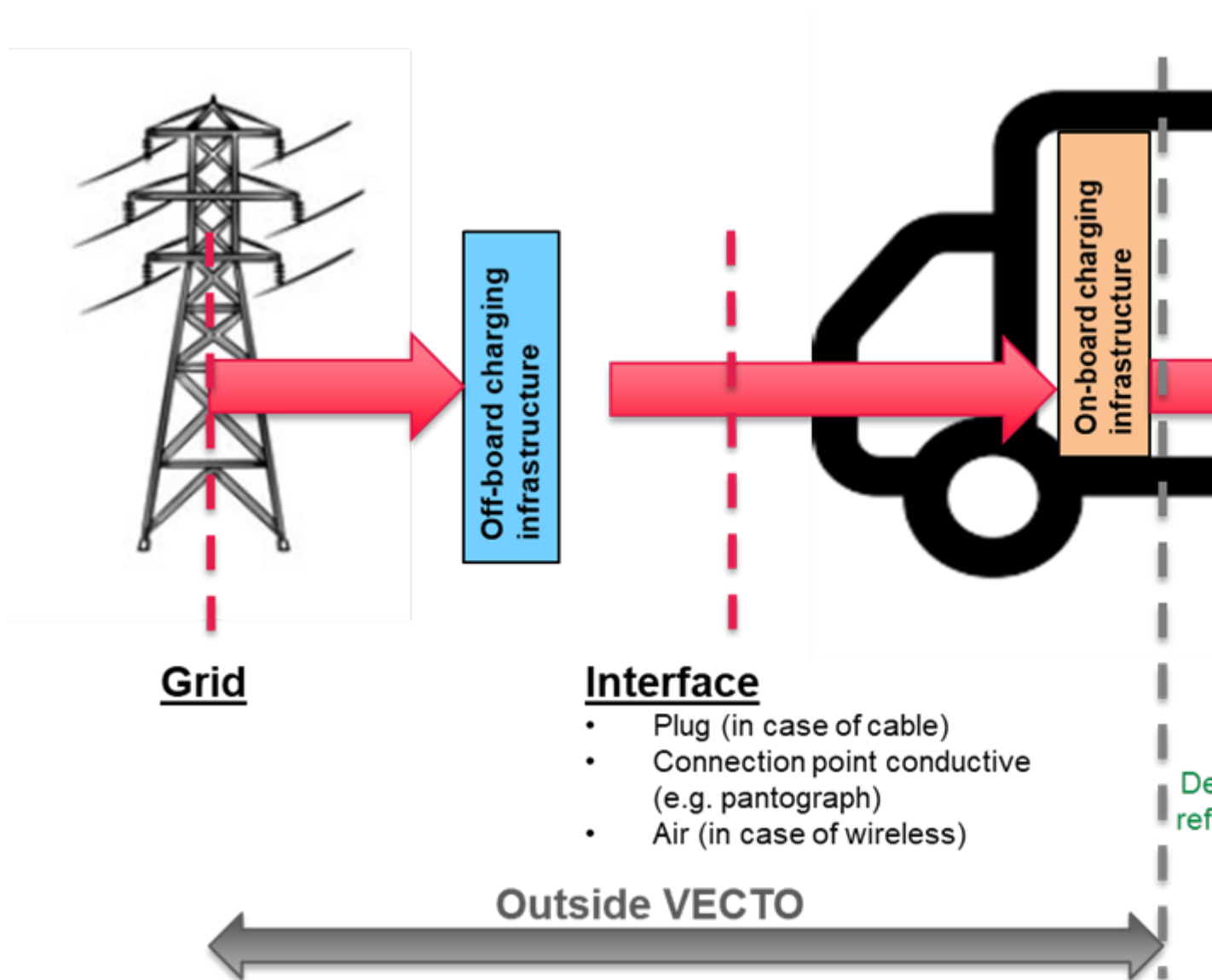
First VECTO beta release for the 3rd Amendment Vehicles with In-Motion Charging Features

Vehicles with In-Motion Charging (IMC) features (1/11)

- One of the tasks of the pre-development project for the 3rd Amendment was to analyze different systems for in-use charging of electric vehicles
 - State of the art
 - Relevance and possible paths of implementation to Regulation (EU) 2017/2400
 - Functional prototype
- The analyses and proposals for (1) and (2) were presented and discussed with stakeholders in five stakeholder meetings (Oct 2021, Dec 2021, Jan 2022, Jun 2022 and Sept 2022).
- The following slides summarize the most important principles and present the functional prototype

Reference point for results regarding electric energy consumption

- Principle applied already for the 2nd amendment
- Consequences for the 3rd amendment
 - Stationary charging: minor improvements to the Annex (max. charging power)
 - In-motion charging: Requires model extension and separate provisions in the Annex

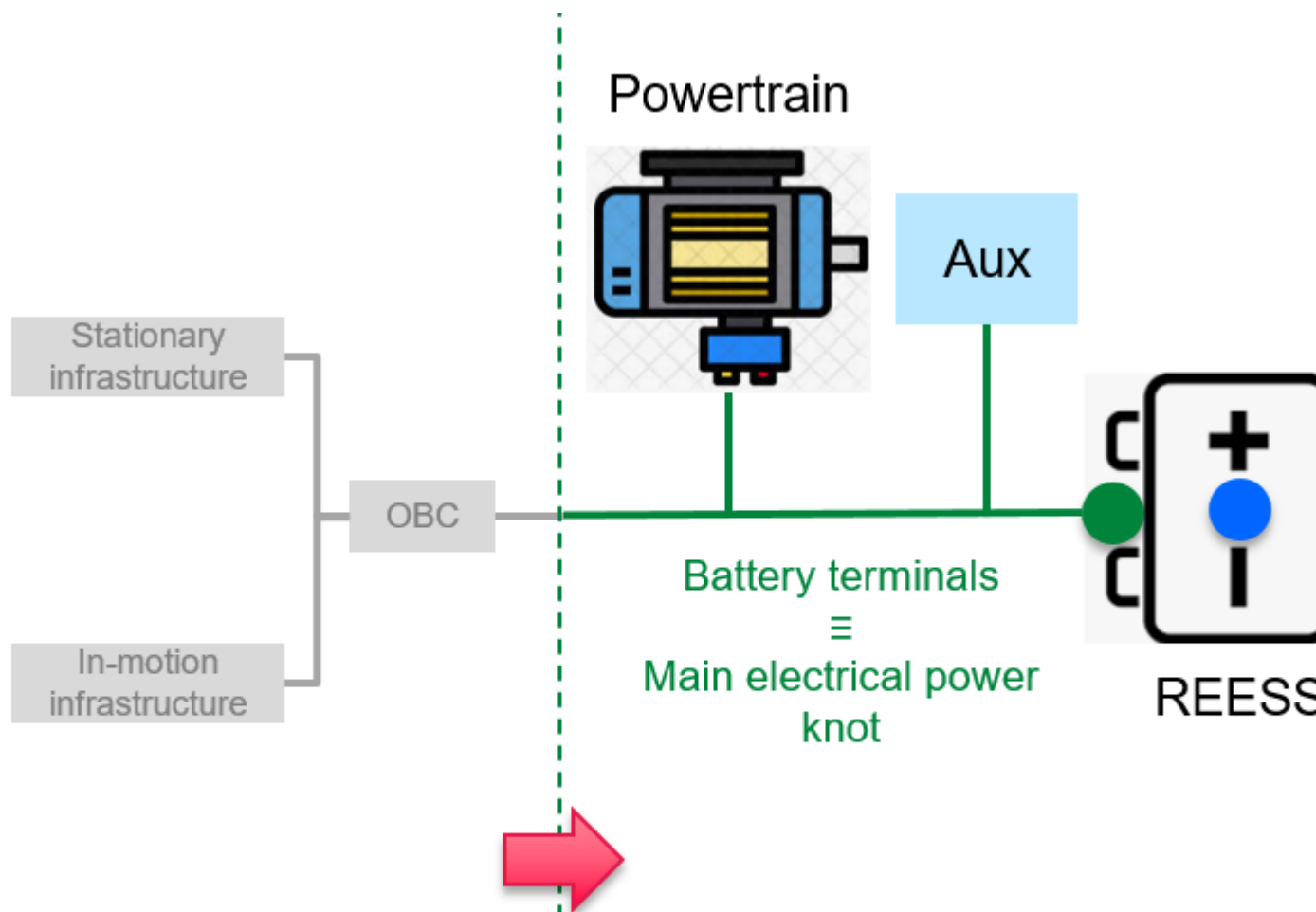


- Modelling in VECTO - Overview
 - *Driving with "direct feed"* incorporated to simulations of "charge depleting" mode by postprocessing (no additional sim time, no additional definitions etc.) (DECL) higher utility factor / share of electric driving for OVC-HEV (slightly) lower battery internal losses.
 - *Charging of batteries by IMC* handled by postprocessing (DECL) higher utility factor / share of electric driving for OVC-HEV.
 - *Impact on air drag* incorporated to in-the loop simulation via a mission average share "active" (which applies either to motorway sections or to the total cycle) (ENG, DECL)
 - *No direct influence on the "electric ranges"*, i.e. these are also only calculated on the basis of the usable battery capacity (DECL)

DECL / ENG: Effect visible in Declaration Mode / Engineering Mode

Report with detailed documentation to be published soon

Lila terms: new elements introduced for IMC modelling



The electric energy crossing this system boundary shall be reflected in the results

$$EC_{el,terminal} = EC_{el,SOC} \cdot \left(S_{bat} \cdot \frac{1}{\eta_{bat(1)}} + S_{dir} \cdot \eta_{bat(2)} \right)$$

Where:

S_{bat} Share of electric energy from grid first charged into battery

S_{dir} Share of electric energy from grid used for direct feed

$\eta_{bat(1)}$... Eta_battery, average charging efficiency when charging from external sources

$\eta_{bat(2)}$... Eta_battery, average discharging efficiency during driving

Calculation examples see *Task3_Masterexcel_OVC_IMC.xlsx* as distributed with ppt Report with detailed documentation to be published soon

In-motion charging “technology”	Relevance			Proposal for definition in Regulation (EU) 2017/2400
	Heavy lorries	Heavy buses	Medium lorries	
Overhead pantograph	Yes	Yes	No	Vehicle equipped with overhead pantograph for connection with overhead catenary infrastructure as regulated by standard tbd and which is not overhead trolley.
Overhead trolley	No	Yes	No	Vehicle equipped with poles for connection with overhead catenary infrastructure as regulated in Annex 12 to UN Regulation 107 revision 8.
Ground-rail	Yes1	Yes1	Yes1	Charging technology that conductively transfers the electrical energy to the vehicle through rails embedded in or on top of the road surface.
Wireless	Yes1	Yes1	Yes1	Charging technology that inductively transfers the electrical energy to the vehicle through devices embedded in or on top of the road surface providing magnetic fields as they are specified IEC 61980.

1. Only Potential niche applications are conceivable.

Lila: to be worked out for the 3 rd _ amendment _

Modelling in VECTO - Proposal for IMC infrastructure availability

Overhead - Trolley

Overhead - Pantograph

Vehicle group	Share of distance of total mission where in-motion charging is possible						
	Long haul	Regional delivery	Urban delivery	Municipal	Construction	Heavy Urban	Urban
53	---	0.00	0.00	---	---	---	---
54	---	0.00	0.00	---	---	---	---
1s	---	0.34	0.00	---	---	---	---
1	---	0.34	0.00	---	---	---	---
2	0.50	0.34	0.00	---	---	---	---
3	---	0.34	0.00	---	---	---	---
4	0.50	0.34	0.00	0.00	0.00	---	---
5	0.50	0.34	0.00	---	0.00	---	---
9	0.50	0.34	---	0.00	0.00	---	---
10	0.50	0.34	---	---	0.00	---	---
11	0.50	0.34	---	0.00	0.00	---	---
12	0.50	0.34	---	---	0.00	---	---
16	---	---	---	---	0.00	---	---
31	---	---	---	---	---	0.30	0.30
32	---	---	---	---	---	---	---
33	---	---	---	---	---	0.30	0.30
34	---	---	---	---	---	---	---
35	---	---	---	---	---	0.30	0.30
36	---	---	---	---	---	---	---
37	---	---	---	---	---	0.30	0.30
38	---	---	---	---	---	---	---
39	---	---	---	---	---	0.30	0.30
40	---	---	---	---	---	---	---

Vehicle group	Share of distance of total mission where in-motion charging is available						
	Long haul	Regional delivery	Urban delivery	Municipal	Construction	Heavy Urban	Urban
53	---	0.00	0.00	---	---	---	---
54	---	0.00	0.00	---	---	---	---
1s	---	0.00	0.00	---	---	---	---
1	---	0.00	0.00	---	---	---	---
2	0.00	0.00	0.00	---	---	---	---
3	---	0.00	0.00	---	---	---	---
4	0.00	0.00	0.00	0.00	0.00	---	---
5	0.00	0.00	0.00	---	0.00	---	---
9	0.00	0.00	---	0.00	0.00	---	---
10	0.00	0.00	---	---	0.00	---	---
11	0.00	0.00	---	0.00	0.00	---	---
12	0.00	0.00	---	---	0.00	---	---
16	---	---	---	---	0.00	---	---
31	---	---	---	---	---	0.80	0.80
32	---	---	---	---	---	---	---
33	---	---	---	---	---	0.80	0.80
34	---	---	---	---	---	---	---
35	---	---	---	---	---	0.80	0.80
36	---	---	---	---	---	---	---
37	---	---	---	---	---	0.80	0.80
38	---	---	---	---	---	---	---
39	---	---	---	---	---	0.80	0.80
40	---	---	---	---	---	---	---

Modelling in VECTO - Proposal for IMC infrastructure availability

Ground rail + Wireless

Vehicle group	Share of distance of total mission where in-motion charging is available						
	Long haul	Regional delivery	Urban delivery	Municipal	Construction	Heavy Urban	Urban industrial
53	---	0.00	0.50	---	---	---	---
54	---	0.00	0.50	---	---	---	---
1s	---	0.00	0.50	---	---	---	---
1	---	0.00	0.50	---	---	---	---
2	0.00	0.00	0.50	---	---	---	---
3	---	0.00	0.50	---	---	---	---
4	0.00	0.00	0.50	0.50	0.00	---	---
5	0.00	0.00	0.50	---	0.00	---	---
9	0.00	0.00	---	0.50	0.00	---	---
10	0.00	0.00	---	---	0.00	---	---
11	0.00	0.00	---	0.50	0.00	---	---
12	0.00	0.00	---	---	0.00	---	---
16	---	---	---	---	0.00	---	---
31	---	---	---	---	---	0.50	0.50
32	---	---	---	---	---	---	---
33	---	---	---	---	---	0.50	0.50
34	---	---	---	---	---	---	---
35	---	---	---	---	---	0.50	0.50
36	---	---	---	---	---	---	---
37	---	---	---	---	---	0.50	0.50
38	---	---	---	---	---	---	---
39	---	---	---	---	---	0.50	0.50
40	---	---	---	---	---	---	---

- Modelling in VECTO - Further proposed assumptions
 - Battery charging during mission
 - 50% of usable battery capacity via “SOC lift” by IMC
 - No further stationary charging
 - Air drag
 - Considered only for “Overhead - pantograph”
 - $\Delta C_{dx}A = 0.6 \text{ m}^2$ for the shares in the mission where IMC is available

Inputs in Graphical User Interface (ENG)

VEH Editor

Parallel Hybrid Vehicle

Tractor Undefin

Technically Permissible Maximum Laden Mass [t]

HDV Group N3

General Powertrain Electric Machine REESS In Motion Charging Torque Limits ADAS PTO

☒ Has In Motion Charging

Share of in motion charging infrastructure availability on total mission distance [%]

Delta CdxA with active in-motion charging feature [m²]

In-motion charging feature only applicable on motorway sections ☐

Attention: In case "In-motion charging feature only applicable on motorway sections" is selected, the value for "Share of in-motion charging infrastructure availability on total mission distance" must be aligned with the distance shares declared as "highway" in the cycle (.vdri).

New tab in "VEH Editor" for HEV and PEV vehicles

Attention!

Working with ENG mode simulations for IMC features requires detailed knowledge of the proposed approach (mainly for developers)

For checking of regulatory impact DECL mode testing recommended!

Inputs in Graphical User Interface (DECL)

VEH Editor

Parallel Hybrid Vehicle

Tractor 4x2

Technically Permissible Maximum Laden Mass [t]

HDV Group N3

General Powertrain Electric Machine REESS In Motion Charging Torque Limits ADAS PTO

In-motion charging technology: Overhead Pantograph

- Dropdown Menu with pre-defined technologies
 - None

- Overhead Pantograph
- Overhead Trolley
- Ground Rail
- Wireless
- Example vehicle models will be distributed with the release (located in “\Generic vehicles”)
- Base specs example models
 - Group 5 / HEV P2 / OVC: P_cont_EM: 200kW | E_Reess: 120kWh nominal
 - Group 5 / PEV: P cont_EM : 325kW | E Reess : 170kWh nominal
- Example results: Long haul with rep payload Comparison results “w/o” and “w/” pantograph with similar other vehicle specs like mass
 - Group 5 / P2 / OVC w/o pantograph: UF = 0.149; CO2 weighted = 692.2 g/km
 - Group 5 / P2 / OVC w/ pantograph: UF = 0.660; CO2_weighted = 347.1 g/km -> -50% CO2
 - Group 5 / PEV w/o pantograph: EC_el = 1.262 kWh/km
 - Group 5 / PEV w/ pantograph: EC_el = 1.297 kWh/km -> +2.8% EC_el